

Credit - Drishti Ias

Nuclear Power on the Moon



For Prelims: NASA , Nuclear reactor , Moon's south pole , Radioisotope Thermoelectric Generators , 1967 Outer Space Treaty (OST) , Artemis Accords

For Mains: International governance of lunar resources and infrastructure, Geopolitical competition over the lunar south pole and water-ice resources.

Source: TH

Why in News?

NASA, in collaboration with the US Department of Energy (DOE), is accelerating its **Fission Surface Power (FSP) project** to deploy a **nuclear reactor** near the **Moon's south pole** by 2030. Simultaneously, China and Russia have announced plans to install an automated nuclear power plant by 2035 for their **International Lunar Research Station (ILRS)**.

- This shift toward industrial-scale space operations raises fresh legal, environmental, and geopolitical questions over lunar territory, resource control, and international governance.

Summary

- Nuclear power is emerging as a critical technology for sustained lunar exploration, offering reliable energy during the long lunar night and enabling activities such as ISRU, life support and resource extraction.
- Its deployment also raises legal, environmental and geopolitical concerns, requiring stronger international space governance, nuclear safety standards and multilateral rules through platforms such as UN-COPUOS, with India playing an active role.

What is the Fission Surface Power (FSP) Project?

- **Core Objective:** In January 2026, NASA and the US DOE signed a Memorandum of Understanding (MoU) to strengthen cooperation on developing, fuelling, authorising and preparing a lunar fission surface

power reactor for launch by 2030 under the [Artemis programme](#).

- **Technical Specifications:** It is designed to be lightweight (under 6,000 kg) and utilizes High Assay Low Enriched Uranium (HALEU) as fuel.
- **Operational Role:** It will provide baseload power for lunar habitats, life-support systems, rovers, and **in-situ resource utilization (ISRU) machinery**, serving as a critical stepping stone for future crewed missions to Mars.
- **Earlier Space Reactor:** The US launched **SNAP-10A**, its first space nuclear fission reactor, in April 1965. It operated in Earth orbit for 43 days, generating approximately 500–600 watts of electricity before shutting down due to an electrical system malfunction unrelated to the reactor.

Note: Nuclear electricity was first generated on the lunar surface during **NASA's Apollo 12 mission (1969)**, which deployed the plutonium-238-powered SNAP-27 Radioisotope Thermoelectric Generator (RTG), designed to produce at least 63 watts of electricity for scientific instruments.

- Unlike a nuclear fission reactor, SNAP-27 generated electricity directly from the heat released by radioactive decay, without a controlled nuclear chain reaction.

Why is Nuclear Power Essential for Lunar Exploration?

- **Overcoming the Lunar Night:** A single day-night cycle on the **Moon lasts 28 Earth days**, meaning bases will endure roughly **14 continuous days of frigid darkness**.
 - Solar arrays cannot sustain life-support systems during this extended nocturnal window without **massively impractical battery reserves**.
- **Accessing Permanently Shadowed Regions (PSRs):** Lunar exploration heavily targets the **south pole**, where deep craters remain in perpetual darkness.
 - These craters host deposits of water ice essential for human survival and fuel production, yet they receive zero sunlight, rendering solar arrays unviable.
- **Scaling Beyond RTGs:** Traditional deep-space probes use [Radioisotope Thermoelectric Generators \(RTGs\)](#), which convert the decay heat of plutonium-238 into electricity.
 - However, RTGs produce only a few hundred watts. Human bases require tens to hundreds of kilowatts, a scale only compact fission reactors can currently provide.
- **Enabling In-Situ Resource Utilisation (ISRU):** Converting lunar ice into breathable oxygen and liquid hydrogen propellants requires industrial power.
 - Producing 10 tonnes of oxygen via ice extraction, electrolysis and hydrogen liquefaction requires an estimated 68 kW of continuous electricity, making **compact fission reactors the only feasible stepping stone to sustainable**, off-Earth propellant depots.

What are the Geopolitics of Nuclear Deployment on the Moon?

- **Scarcity of Prime Lunar Real Estate:** While the Moon is vast, viable base locations near the south pole that offer accessible water-ice, safe landing terrain, and Earth communication lines **are extremely limited**.

- **Infrastructure as Influence:** The **1967 Outer Space Treaty (OST)** expressly forbids national appropriation or sovereign claims over celestial bodies.
 - However, deploying a reactor **establishes de facto control over an area** .
- **The "Safety Zone" Loophole:** NASA design studies have examined locating reactors approximately 1 km away from users, although this is not a universal international requirement.
 - The **US-led Artemis Accords** allow for temporary **"safety zones" to prevent harmful interference**.
- **Infrastructure as Influence:** Deploying reactors, power cables and associated infrastructure near **valuable lunar resources** could provide early-operating nations with strategic advantages.
 - **Article IX of the Outer Space Treaty** mandates due regard for other states' interests and consultations to prevent harmful interference, but does not grant reactor operators exclusive territorial rights.

India's Position

- **ISRO and the Department of Atomic Energy (DAE)** are collaborating to develop **indigenous nuclear power technologies for space exploration** , strengthening India's strategic autonomy.
- To support future long-duration missions like the **Bharatiya Antariksh Station and the 2040 crewed lunar landing** , ISRO is exploring 100-watt Radioisotope Thermoelectric Generators (RTGs).
 - This builds on the success of **Chandrayaan-3**, which carried two 1-watt Radioisotope Heater Units (RHUs) exclusively on its propulsion module.
 - Crucially, while the RHUs used on Chandrayaan-3 only supplied thermal heat to prevent freezing, the proposed RTGs will convert nuclear decay heat directly into electrical power.

What are the Concerns Regarding the Nuclear Deployment Over the Moon?

Technical and Environmental Hurdles

- **Mass and Shielding Logistics:** Protecting astronauts and surface equipment from reactor radiation requires **heavy shielding**.
- **Power Transmission Challenges:** Locating reactors away from human habitats to minimise radiation exposure creates additional infrastructure requirements, including robotic deployment systems, long-distance power cables and reliable electrical distribution networks across difficult lunar terrain.
- **Heat Dissipation in a Vacuum:** Lunar reactors generate substantial waste heat. Because the Moon lacks an atmosphere to facilitate **convection, systems** must rely on massive radiator panels.
- **Autonomous Operation and Reliability:** Lunar reactors must operate autonomously for several years with **minimal human intervention** , adjusting electricity generation to changing demand while withstanding launch vibrations, extreme temperatures and limited opportunities for maintenance.
- **Launch Safety and Core Activation:** Space reactors must be launched with unirradiated, **"cold" fuel cores to prevent radiological contamination in the event of a terrestrial launch vehicle failure**.
- **Risk of Irreversible Contamination** : A mechanical failure, reactor breach, or cooling loss could release radioactive material into the few pristine, scientifically invaluable crater basins that contain usable water ice.

Governance and Regulatory Shortfalls

- **Non-Binding U.N. Principles** : The **1992 UN Principles Relevant to the Use of Nuclear Power Sources in Outer Space (UNGA Res 47/68)** is a non-binding resolution that provides guidelines for safety and notification, **but lacks enforcement authority**.
- **Omission of Advanced Propulsion**: The **1992 UN Principles** apply strictly to power sources generating electricity (RTGs and basic fission reactors) and do not address modern **Nuclear Thermal Propulsion (NTP) or Nuclear Electric Propulsion (NEP)**.
- **Absence of End-of-Life and Disposal Protocols** : Current space treaties provide no clear, binding standards for decommissioning spent lunar reactors, containing irradiated waste, or disposing of jettisoned orbital reactors.
- **Ambiguities in Liability and Environmental Law**: The **1972 Liability Convention** governs damage caused by space objects on Earth or to aircraft, but offers ambiguous guidance regarding **nuclear accidents and radioactive contamination in cis-lunar space or on celestial surfaces**.
- **Tension Over "Freedom of Access"**: International space law guarantees the right of all states to freely explore celestial bodies.
 - Unilateral declarations of reactor safety perimeters create friction between legitimate operational safety and the prohibition against monopolizing strategic lunar corridors.

What Measures are Needed for Space Governance and India?

- **Modernize the 1992 UN Framework**: The **UN Committee on the Peaceful Uses of Outer Space (COPUOS)** must draft binding technical standards covering high-output surface fission plants, advanced propulsion systems, and end-of-life radioactive waste disposal.
- **Establish an IAEA for Space**: The international community should create a multilateral oversight body modeled after the **International Atomic Energy Agency (IAEA)** to certify space reactor designs, monitor space-bound fissile material, and verify compliance.
- **Multilateral Consensus on Safety Zones**: Safety perimeters must be defined through international consultation rather than unilateral declarations. This ensures zones remain strictly functional, temporary, and non-exclusionary.
- **Enhance ISRO-DAE Synergy**: India must aggressively **fund the domestic development of space-grade nuclear reactors** to secure a leadership role in deep-space exploration and gain a stronger negotiating position in global space governance.

Conclusion

As humanity moves towards **sustained lunar settlements** , nuclear energy could become a key source of reliable space power. However, the use of **nuclear materials in space** requires stringent safety standards to prevent risks to Earth and the lunar environment. The international community should strengthen **global nuclear safety norms through UN-COPUOS** to ensure the responsible and sustainable use of nuclear technology in space.

Drishti Mains Question:

In-Situ Resource Utilisation could transform the Moon from a destination into an infrastructure hub for deep-space exploration. Discuss

Frequently Asked Questions

1. What is NASA's Fission Surface Power Project?

The **Fission Surface Power (FSP) Project** aims to develop a compact **nuclear fission reactor for lunar surface power** , targeting around **40 kWe** , with designs exploring higher capacities.

2. Why is nuclear power important for lunar exploration?

Nuclear reactors can provide **continuous baseload power** during the Moon's approximately **14-day lunar night** and in **Permanently Shadowed Regions** , where solar energy is unavailable.

3. What does the Outer Space Treaty say about lunar resources and territory?

The **1967 Outer Space Treaty** prohibits national appropriation of celestial bodies. However, the growing use of permanent infrastructure raises questions about **de facto control, access and resource governance** .

4. What are the UN Principles on Nuclear Power Sources in Outer Space?

The **1992 UN Principles (UNGA Resolution 47/68)** provide safety and notification guidelines for nuclear power sources in space but are **non-binding** and lack comprehensive rules for emerging high-power lunar reactors.

5. What role can India play in lunar nuclear power and space governance?

India can strengthen **ISRO-DAE cooperation** , develop indigenous space nuclear technologies and use platforms such as **UN-COPUOS** to promote **multilateral safety standards, equitable access and responsible lunar exploration** .

[Watch Video on YouTube:

▶ <https://www.youtube.com/embed/RjNfAIP1ko>

]

UPSC Civil Services Examination Previous Year Question (PYQ)

Prelims

Q. Consider the following statements: (2016)

The Mangalyaan launched by ISRO

1. is also called the Mars Orbiter Mission
2. made India the second country to have a spacecraft orbit the Mars after USA
3. made India the only country to be successful in making its spacecraft orbit the Mars in its very first attempt

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Ans: (c)

Q.2 What is the purpose of the US Space Agency's Themis Mission, which was recently in the news? (2008)

- (a) To study the possibility of life on Mars
- (b) To study the satellites of Saturn
- (c) To study the colourful display of high latitude skies
- (d) To build a space laboratory to study the stellar explosions

Ans: (c)

Mains

Q. What is the main task of India's third moon mission which could not be achieved in its earlier mission? List the countries that have achieved this task. Introduce the subsystems in the spacecraft launched and explain the role of the 'Virtual Launch Control Centre' at the Vikram Sarabhai Space Centre which contributed to the successful launch from Sriharikota. **(2023)**

Q. What is India's plan to have its own space station and how will it benefit our space programme? **(2019)**

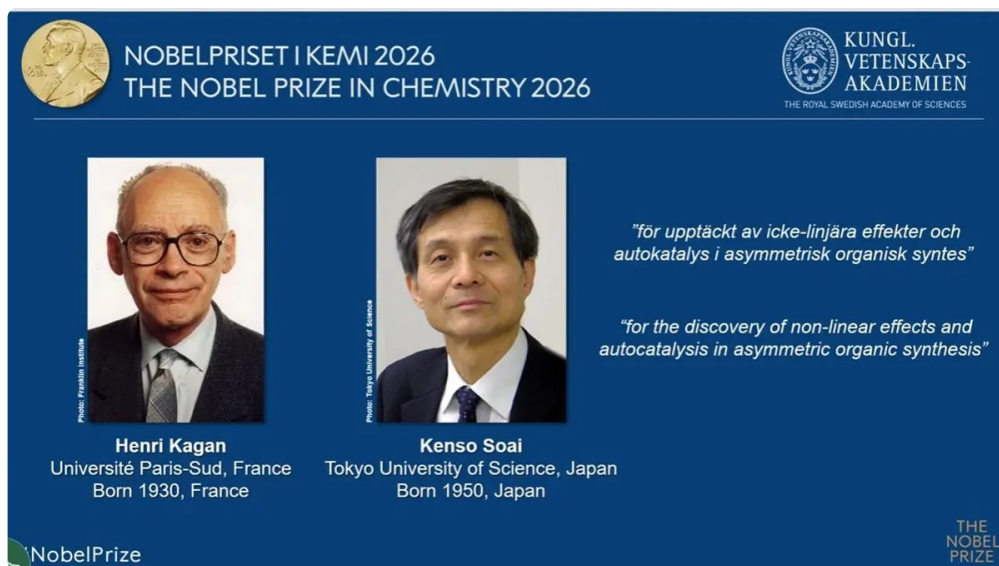
Nobel Prize in Chemistry 2026: Non-Linear Effects & Autocatalysis



Why in News?

The Royal Swedish Academy of Sciences has awarded the **Nobel Prize in Chemistry 2026** to **Henri B. Kagan (Université Paris-Sud, France)** and **Kenso Soai (Tokyo University of Science, Japan)**. They were honoured for their groundbreaking **“discovery of non-linear effects and autocatalysis in asymmetric organic synthesis”**.

- Their foundational work explains how chemical reactions can amplify small asymmetries in molecules, solving a century-old mystery regarding how biological life came to overwhelmingly favor specific molecular structures.



What is Molecular Asymmetry or Chirality?

- **Chiral Molecules:** Many fundamental molecules, including **amino acids** (the building blocks of proteins), exist in two forms that are **non-identical mirror images** of each other. This property is known as **chirality, derived from the Greek word for "hand"**.
- **Left-Handed and Right-Handed Variants:** Just like human hands, these molecules cannot be perfectly superimposed on one another. They are often referred to as **left-handed and right-handed variants (enantiomers)**.
- **Homochirality in Life:** While the two variants are chemically identical in a symmetrical environment, living organisms **overwhelmingly favour and use only one form**.
 - For instance, the proteins in human cells are exclusively made up of **"left-handed" amino acids**.
- **The Century-Old Puzzle:** Traditional chemical syntheses in the laboratory typically produce equal amounts of both mirror-image molecules (known as a racemic mixture).
 - For decades, scientists struggled to understand how biological life evolved to be strictly "homochiral" (having molecules of only one of the two possible mirror-image forms).

What are the Key Facts About the Nobel Prize in Chemistry 2026?

- **Breakthrough:** Before Kagan's research, chemists operated on a strict assumption: they believed the **purity of a chemical product would always directly mirror** the purity of the catalyst used.
 - If a catalyst mixture was 80% left-handed and 20% right-handed, scientists assumed the resulting product would have that exact same 80/20 ratio (a linear relationship)
 - The 2026 Nobel laureates discovered chemical mechanisms that can selectively drive reactions to produce exclusively one of the two possible mirror images.
- **Henri Kagan's Contribution (1986):** **Henri Kagan** discovered that **chiral catalysts can produce a much greater chiral imbalance than expected** .
 - When catalyst molecules pair up, **opposite-handed pairs are less active** , suppressing the minority form. This allows the **majority-handed catalyst to dominate the reaction** , producing a product with a **disproportionately high excess of one mirror-image molecule** .
 - This phenomenon, known as the **non-linear effect** , became an important clue in understanding **chiral amplification and biological homochirality** .
- **Kenso Soai's Autocatalysis (1995-2003):** Building on **Kagan's work** , Kenso Soai demonstrated that a tiny initial chiral imbalance could be dramatically amplified through **autocatalysis** , where a chemical product catalyses the formation of more of itself.
 - **The Soai Reaction (1995):** Soai discovered a **chiral 5-pyrimidyl alkanol** that could act as a catalyst for its own formation. This was a major breakthrough in **asymmetric synthesis** .
 - **The Snowball Effect (2003):** Soai showed that even a **very small initial excess of one mirror-image molecule** could be dramatically amplified. The dominant form catalyses the formation of **more molecules of the same handedness** , causing the imbalance to grow with each reaction cycle.
 - This **self-amplifying process** can drive a mixture towards **very high enantiomeric purity** , providing an important experimental model for understanding how **biological homochirality** may have emerged from a tiny initial molecular asymmetry.

Note: In **1953** , physicist **Charles Frank** proposed that biological **homochirality** could emerge through three conditions:

- **A** chiral catalyst drives an asymmetric reaction.
- One enantiomer is amplified while the other is suppressed.
- The product acts as its own catalyst through **autocatalysis** , creating a self-reinforcing cycle.
- **Kagan's non-linear effects** supported the second condition, while **Soai's asymmetric autocatalysis** demonstrated the third, providing an experimental basis for understanding the possible emergence of homochirality.

Significance

- **Pharmaceutical Manufacturing:** Many drugs exist as **enantiomers** , where one form may provide the desired therapeutic effect while the other may have reduced or harmful effects. Kagan and Soai's work has strengthened the foundations of **asymmetric synthesis** , enabling chemists to selectively produce the desired enantiomer.
- **Origin of Life:** Their work offers a possible chemical explanation for **biological homochirality** . It shows how a tiny initial molecular imbalance could be **amplified into a strong preference for one molecular handedness** .

- **Materials Science: Chiral control** is increasingly important in designing advanced materials with specific molecular, optical, and electronic properties.
- **Agrochemicals and Fragrances: Enantioselective synthesis** helps produce specific enantiomers of pesticides, fragrances, and other specialty chemicals, improving **efficiency, selectivity, and performance** .

Nobel Prize 2026

- [Nobel Prize in Physics 2026: IceCube Neutrino Observatory](#)
- [2026 Nobel Prize in Physiology or Medicine: Optogenetics](#)

Frequently Asked Questions

1. Who won the Nobel Prize in Chemistry 2026?

The **2026 Nobel Prize in Chemistry** was awarded to **Henri B. Kagan** of France and **Kenso Soai** of Japan for their work on **non-linear effects and autocatalysis in asymmetric organic synthesis** .

2. What is chirality?

Chirality is the property of molecules that exist as **non-superimposable mirror images** , known as **enantiomers** or left- and right-handed forms.

3. What is the non-linear effect discovered by Henri Kagan?

It occurs when a small difference in **catalyst chirality produces a disproportionately large difference in product chirality** , resulting in **chiral amplification** .

4. What is autocatalysis demonstrated by Kenso Soai?

Autocatalysis occurs when a chemical product catalyses its own formation. The **Soai reaction** demonstrated how a tiny initial chiral imbalance can be strongly amplified.

5. Why is the work of Kagan and Soai significant?

Their discoveries provide insights into **biological homochirality** and have applications in **pharmaceuticals, agrochemicals, fragrances and advanced materials** .

[Watch Video on YouTube:

▶ <https://www.youtube.com/embed/nw8vT7X4AEo>

]

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims

Q. What is blue carbon? (2021)

- (a) Carbon captured by oceans and coastal ecosystems
- (b) Carbon sequestered in forest biomass and agricultural soils
- (c) Carbon contained in petroleum and natural gas
- (d) Carbon present in atmosphere

Ans: (a)

Q. Bisphenol A (BPA), a cause of concern, is a structural/key component in the manufacture of which of the following kinds of plastics? (2021)

- a) Low-density polyethylene
- b) Polycarbonate
- c) Polyethylene terephthalate
- d) Polyvinyl chloride

Ans: (b)

Reconstitution of Department-Related Parliamentary Standing Committees



Source: TH

Why in News?

Recently, the **Lok Sabha Speaker and Rajya Sabha Chairman** reconstituted various **Department-related Parliamentary Standing Committees** for 2026-27.

What are the Key Facts About Department-Related Parliamentary Standing Committees?

- **About: Department-Related Standing Committees (DRSCs)** are Parliamentary committees that enable detailed scrutiny of the **functioning, finances, legislation and policies of Union Ministries and Departments**, thereby strengthening executive accountability to Parliament.

- **Number of Committees:** The present system of DRSCs was **established in 1993**. The system was expanded from **17 to 24 DRSCs in 2004** , **covering all Ministries and Departments** of the Union Government.
- **Composition:** Each **DRSC consists of 31 Members—21 from the Lok Sabha and 10 from the Rajya Sabha** . Representation of parties and groups is broadly proportional to their strength in the two Houses.
- **Ministers Excluded:** A **Minister cannot be nominated as a member of a DRSC**. If a member becomes a Minister after nomination, he or she ceases to be a member of the Committee.
- **Chairperson:** **Chairpersons of committees listed under Part I of the Fifth Schedule** are appointed by the Chairman of the Rajya Sabha, while those under **Part II are appointed by the Speaker of the Lok Sabha** from among Committee members.
- **Tenure:** Members of DRSCs are appointed for a **term not exceeding one year**, and the Committees are reconstituted annually.
- **Major Functions :** DRSCs examine **Demands for Grants of concerned Ministries, Bills referred to them, Annual Reports of Ministries/Departments, and National Basic Long-Term Policy Documents** referred by the Presiding Officers.
- **Limits on Jurisdiction:** They **do not examine day-to-day administration of Ministries and generally avoid matters already under consideration** by other Parliamentary Committees.
- **Nature of Recommendations:** Their reports are **advisory in nature and have persuasive value**; they are treated as considered advice of the Committee rather than binding decisions.
- **Quorum and Proceedings :** The **quorum is approximately one-third of the total membership; for a 31-member DRSC, the quorum is 10** . Committee sittings are held in private and are not open to the media.
- **Consensus-Based Reports:** DRSC reports are **generally based on broad consensus, though members may submit a note of dissent**, which is presented to the House along with the report.
- **Expert Inputs:** Committees **may call for papers, records and witnesses**, and may also seek expert or public opinion while examining a subject.

Departmental Standing Committee

The committee is to secure more accountability of the Executive (that is Council of Ministers) to the Parliament, particularly financial accountability.

24 Standing committees each consists of
31 members



21 from Lok Sabha (LS)



Members of LS are nominated by the **Speaker** from amongst its members.



10 from Rajya Sabha (RS)



Members of RS are nominated by the **Chairman** from amongst its members.

- A **Minister** is not eligible to be nominated as a member of any standing committees.
- In the case of a member, after his nomination to any standing committee, is appointed as a minister, he then **ceases** to be a member of committee.

Out of 24 Standing Committee

16 are under Lok Sabha



8 are under Rajya Sabha



Parliamentary Committees

- **About:** A **Parliamentary Committee** is a body constituted by the **Lok Sabha or Rajya Sabha** , or nominated by the **Speaker/Chairman** , to perform functions delegated by Parliament.
- Parliamentary Committees:
 - Work under the direction of the **presiding officer** .
 - Submit their reports to the **House or the Speaker/Chairman** .
 - Are serviced by the **Lok Sabha/Rajya Sabha Secretariat** .
- The committee system, inspired by the **British Parliament** , derives its constitutional basis mainly from **Article 105** , relating to parliamentary powers and privileges, and **Article 118** , relating to regulation of parliamentary business.

Frequently Asked Questions (FAQs)

1. What are Department-Related Parliamentary Standing Committees (DRSCs)?

DRSCs are parliamentary committees established in 1993 to examine Ministry-wise expenditure, Bills, policy issues and implementation of government programmes.

2. How many Department-Related Parliamentary Standing Committees are there?

There are 24 DRSCs, each comprising 31 members — 21 from Lok Sabha and 10 from Rajya Sabha.

3. What is the tenure of Department-Related Parliamentary Standing Committees?

The tenure of each DRSC is one year, after which it is reconstituted.

4. Who reconstitutes Department-Related Parliamentary Standing Committees?

The Lok Sabha Speaker and Rajya Sabha Chairman reconstitute the committees under their respective jurisdictions.

5. What are the major functions of DRSCs?

DRSCs examine Demands for Grants, scrutinise referred Bills, review government policies and programmes, and submit reports and recommendations to Parliament.

[Watch Video on YouTube:

▶ <https://www.youtube.com/embed/Ou21BbeMJjw>

]

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims

Q. With reference to the Parliament of India, which of the following Parliamentary Committees scrutinizes and reports to the House whether the powers to make regulations, rules, sub-rules, by-laws, etc., conferred by the Constitution or delegated by the Parliament are being properly exercised by the Executive within the scope of such delegation? (2018)

- (a) Committee on Government Assurances
- (b) Committee on Subordinate Legislation
- (c) Rules Committee
- (d) Business Advisory Committee

Ans: (b)

Q. Consider the following statements: (2013)

1. The Parliamentary Committee on Public Accounts
2. consists of not more than 25 members of the Lok Sabha.
3. scrutinizes appropriation and finance accounts of the Government.
4. examines the report of the Comptroller and Auditor General of India.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 and 3 only
- c) 3 only
- d) 1, 2 and 3

Ans: (b)

Q. Which one of the following is the largest Committee of the Parliament? (2014)

- (a) The Committee on Public Accounts
- (b) The Committee on Estimates
- (c) The Committee on Public Undertakings
- (d) The Committee on Petitions

Ans: (b)

Mains

Q. Do Department-related Parliamentary Standing Committees keep the administrations on their toes and inspire reverence for parliamentary control? Evaluate the working of such committees with suitable examples. (2021)

Recapture of Kidal in Mali



Source: TH

The Malian Armed Forces (FAMA), backed by Russia's Africa Corps, have regained control of the strategic northern city of Kidal after the Tuareg-led Azawad Liberation Front (FLA) announced a controlled withdrawal. The development is a symbolic boost for **Mali's** military government, though it does not mark a decisive end to the wider insurgency.

- **Geographical & Strategic Significance** : Located in the **Sahel region** near the **Algerian border** , **Kidal** is a highly strategic desert transit hub.
 - It serves as a major political and symbolic centre of the Tuareg separatist movement and the idea of 'Azawad' (the name used by the ethnic **Tuareg** population for their desired independent state in northern **Mali**) .
- **Symbol of State Sovereignty** : For the central military government based in **Bamako** , recapturing Kidal restores a crucial symbol of state sovereignty over its northern peripheries.
 - Its rugged **Adrar des Ifoghas** mountain massif has historically harbored insurgents and provided immense leverage over cross-border trade and smuggling networks.

Mali

- **Geography**: Mali is bordered by seven nations: **Algeria** to the north, **Niger and Burkina Faso** to the east, **Côte d'Ivoire and Guinea** to the south, and **Senegal and Mauritania** to the west.
- **Relief and Prominent Landforms**: Notable topographic features include the **Mandingue Plateau** , the **Bandiagara Escarpment (Dogon Plateau)** , the **Iforas Massif (Adrar des Ifoghas)** in the northeast, and **Mount Hombori Tondo (1,155 m)** , which is the **highest point in Mali** .
- **River**: The **Niger River** runs through Mali, acting as its primary economic artery.
- **Historical Civilizations**: Historically part of the precolonial **Ghana, Mali, and Songhai empires** , the country hosts renowned cultural hubs such as **Timbuktu** (trans-Saharan trade crossroad), **Djenné** (Sudano-Sahelian architecture and Great Mosque), and the cliffside settlements of the **Bandiagara Escarpment** .
- **Culture**: Major ethnic groups include the Bambara, Fulani, Dogon, and Tuareg.
- **Protected Areas & Conservation**: Major biodiversity reserves include the **Boucle du Baoulé National Park** , the **Ansongo-Ménaka Animal Reserve** , and the **Gourma (Douentza) Elephant Reserve** .



Read more: [Military Coup in Mali](#)

Air Marshal Ashutosh Dixit Appointed 29th Chief of the Air Staff



Source: PIB

Air Marshal **Ashutosh Dixit** has been appointed as the **29th Chief of the Air Staff (CAS)** . He will succeed **Air Chief Marshal Amar Preet Singh** , who will superannuate on **31st October 2026** .

- **Service Profile:** Commissioned into the **Fighter Stream of the Indian Air Force in 1986** he has flown aircraft including **Mirage-2000 , MiG variants , Jaguar , IL-78, Hawk and Tejas** . He has nearly **four decades of service, over 3,500 hours of flying experience**, and is an Experimental Test Pilot and Qualified Flying Instructor.
- **Key Appointments:** Before becoming Vice Chief of the Air Staff, he served as **Air Officer Commanding-in-Chief**, Central Air Command, and later as **Chief of Integrated Defence Staff to the Chairman** , Chiefs of Staff Committee (CISC).
- **Operations and Exercises:** He has participated in **Operation Safed Sagar , Operation Rakshak , Exercise Cope India and Operation Sindoor** . He has also commanded a frontline fighter air base in the Western Sector and a premier fighter training base in the Southern Sector.
- **Awards:** He has received the **Vayu Sena Medal (2006), Vishisht Seva Medal (2011), Ati Vishisht Seva Medal (2023)** and **Param Vishisht Seva Medal (2026)** .



Read More: [Operation Sindoor](#)

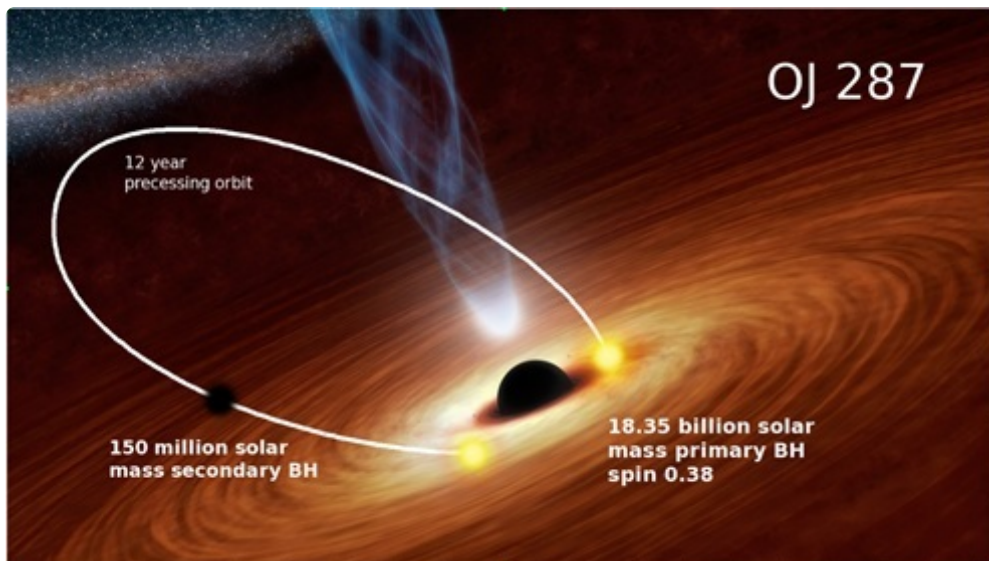
OJ 287: Rare Binary Black Hole System



Source: PIB

A **10-year study of OJ 287** , involving scientists from the **Aryabhata Research Institute of Observational Sciences (ARIES)** , **Nainital, Uttarakhand** , has provided new information about its rare **binary supermassive black hole system** .

- **Key Findings:** The observations helped determine the **relative positions of the two black holes** , study changes in **brightness, spectrum and polarization** , and search for possible **emissions from the smaller black hole** .
 - The central black hole was estimated to have a mass of **at least 3.89 billion times that of the Sun** .
- **About OJ 287:** It is a rare **blazar** located about **4 billion light-years from Earth** , containing **two Supermassive Black Holes (SMBHs)** orbiting each other in a **12-year orbit** .
 - **Blazar:** It produces powerful **plasma jets moving nearly at the speed of light towards Earth** , with emissions ranging from **radio waves to gamma-rays**.
- **Importance:** OJ 287 can help scientists understand **relativistic jets, binary black holes and multimessenger astronomy** , which combines signals such as **light, gravitational waves** , **neutrinos and cosmic rays** .



Read More: [Black Hole](#)

SAREX-2026



Source: PIB

The **12th National Maritime Search and Rescue Exercise (SAREX-2026)** is set to be inaugurated, with a **Search and Rescue (SAR)** demonstration scheduled off the **Paradip coast, Odisha**.

- The exercise is being organised by the **Indian Coast Guard (ICG)** and includes **32 observers from 30 countries**, strengthening international interoperability.
 - The Indian Coast Guard (ICG) is an **Armed Force of the Union** constituted under the **Coast Guard Act, 1978**, responsible for **safeguarding India's maritime interests** and sovereign rights.
 - It acts as the **central coordinating authority for oil-spill response** in India's maritime zones.
- **Participating Forces and Assets:** The exercise will involve the **ICG, Indian Navy and Indian Air Force (IAF)**, deploying **15 ICG ships, 2 Indian Navy ships, 2 Dornier aircraft, 3 Advanced Light Helicopters, 1 Chetak helicopter, 1 C-130J aircraft and Mi-17 helicopters**.
- **Objective:** It will simulate a **Mass Rescue Operation** involving simultaneous aeronautical and maritime emergencies.
 - The scenario includes a passenger aircraft carrying **228 persons ditching at sea and colliding with container vessel MV Zumba**.
- **Significance:** SAREX-2026 supports India's vision of **SAGAR** and **MAHASAGAR** by developing a faster, technology-enabled and internationally coordinated maritime SAR framework.